

Outage Investigation Report —

OIR-2025-005

Field	Value
Document ID	OIR-2025-005
Report Date	2025-01-28
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Reviewing Manager	Priya Ramachandran, Plant Manager, NW-PowerPool
Region	NW-PowerPool
Site	NW-PowerPool-WIND-024
Asset	WIN-10131 (Siemens Gamesa, wind_turbine)
Outage ID	OUT-2025-0572
Classification	Forced outage — overheating
Distribution	NW-PowerPool Operations, Reliability Engineering, OEM Liaison, Compliance

1. Outage Summary

WIN-10131 declared a forced outage on 2025-01-13 at 20:19 PST on a gearbox high-temperature trip. The asset returned to service on 2025-01-19 at 15:19 PST, total duration **139 hours**. Lost generation: **3,389.3 MWh**. This is the first forced-outage event of FY2025 at NW-PowerPool-WIND-024 and the opening event in what later becomes a repeated WIN-10131 overheating pattern (see OIR-2025-002 for the August event).

This report is also the first FY2025 document to flag the conflict between Siemens Gamesa OEM Manual Rev. 4.2 and Cascadia internal Standard MS-2025-03, later formalized in MOC-2025-022.

2. Timeline of Events

Timestamp (PST)	Event
2025-01-13 09:40	Routine SCADA shows main bearing temperature rising 0.5 °C/hr at ambient -4 °C and 7.8 MW load.
2025-01-13 13:55	Bearing crosses 78 °C "monitor" threshold per Siemens Gamesa OEM Manual Rev. 4.2 §7.3.
2025-01-13 17:30	Bearing crosses 82 °C. Under Standard MS-2025-03 §5.1 this should require shutdown; OEM threshold is 88 °C. Operator continued operation per the controlling pre-MOC instruction.
2025-01-13 20:19	Trip on 91 °C oil sump. Forced outage declared.
2025-01-13 22:00	CPS unavailable within 4 hours due to weather (ice on I-84 east of The Dalles). EPA-2024-11 invoked.
2025-01-14 02:30	Apex Emergency Grid Solutions accepts dispatch; arrives 06:45 PST.
2025-01-14 — 2025-01-18	Gearbox oil drain, oil-spectrometer analysis, filter replacement, vibration baseline. Spectrometer reading 14 ppm iron (limit 12).
2025-01-18 — 2025-01-19	Restart sequence, 24-hour surveillance run.
2025-01-19 15:19	Returned to service.

3. Telemetry Observations

- Gearbox oil sump at trip: 91.2 °C.
- Main bearing at trip: 95.8 °C.
- Vibration RMS: 6.9 mm/s on main shaft — above MS-2025-03 6.5 limit, below OEM Manual Rev. 4.2 10.0 threshold.
- Iron particulate: 14 ppm — first exceedance of 12 ppm limit on this unit since commissioning.
- Ambient: -4 to -9 °C. NW wind 12–18 m/s. Conditions within seasonal norm.
- The Q4 2024 quarterly oil-spectrometer test (WO-2024-12118) had been postponed twice and was outstanding at the time of the event; it was finally executed retrospectively during this outage. See CAR-2024-118.

4. Restoration Activities

Vendor	Scope	Invoice (USD)
Apex Emergency Grid Solutions	Onsite diagnostics, oil change, filter set, vibration baseline	\$171,000
Apex / Siemens Gamesa subcontract (1.8x)	OEM warranty review and sign-off	\$98,500
Total		\$269,500

CPS scheduled equivalent estimate (issued 2025-01-22 retrospective): \$71,200.

5. Preliminary Root Cause

Direct cause: oil contamination (iron particulate at 14 ppm) reducing lubrication efficacy under cold-soak conditions.

Contributing causes:

1. Q4 2024 quarterly oil-spectrometer test was deferred under CSM-7. Had it been executed on schedule (December 2024), the particulate trend would have been visible 4–6 weeks earlier.
2. NW-PowerPool entered Q1 2025 with 14 preventive WOs on a deferred list against NW-PowerPool-WIND-014 and -024 — Q1 budget freeze. WO-2025-01044 (this unit's January PM) was among them.
3. Documentation conflict: OEM Manual Rev. 4.2 permits continued operation up to 88 °C bearing; Standard MS-2025-03 requires shutdown ≥ 80 °C sustained > 30 min. The operating staff defaulted to the OEM threshold absent a formal Cascadia override. This is the originating procedural conflict that drives MOC-2025-022.

6. Recommendations

1. Execute the deferred WO-2025-01044 and the remaining 13 Q1 deferred items as the freeze lifts. Track under CAR-2025-002 (Q1 budget freeze recovery plan).
2. Open a Management of Change request to formally override the OEM threshold or formally adopt it as Cascadia's controlling threshold. Tracked as MOC-2025-022 (subsequently opened 2025-03-04).
3. Audit CSM-7 oil-spectrometer compliance across NW-PowerPool. See CAR-2025-006 (later opened).
4. Add WIN-10131 to the high-watch list; this is the first overheating event of the year and the pattern warrants surveillance.

5. Cross-reference CAR-2024-118 (preventive maintenance cadence inadequate for aging wind assets, open since December 2024).

Signatures

Role	Name	Date
Author	Marcus Chen, Sr. Reliability Engineer	2025-01-28
Plant Manager	Priya Ramachandran	2025-01-29
Compliance Review	D. Whitcombe	2025-01-30

Distribution: NW-PowerPool Operations · Reliability Engineering · OEM Liaison · Compliance · CFO Office (cost flag)

References: CAR-2024-118 · CAR-2025-002 · CAR-2025-006 · MOC-2025-022 · EMR-2025-001 · OIR-2025-002 · RCA-2025-001 · RCA-2025-005 · Siemens Gamesa OEM Manual Rev. 4.2 · Standard MS-2025-03 · CSM-7